

Atlas Coaching Intelligence: The Complete Knowledge Architecture

TL;DR

- **Atlas's intelligence should be built as a three-layer brain:** a *Coaching Knowledge Base* (evidence organized as structured rules/tables PLUS prose for retrieval), a *Deterministic Rules Engine* (code that owns every decision where consistency matters — progression, volume math, fatigue/readiness scoring, plateau detection, risk flags), and a *constrained LLM* (which explains, educates, encourages, asks questions, and personalizes language but never overrides the deterministic logic). Build the rules engine first; the LLM is the voice, not the judgment.
- **The evidence is strong on principles but genuinely uncertain on the exact numbers** — total weekly volume drives hypertrophy (Schoenfeld, Ogborn & Krieger 2017 found “each additional set was associated with an increase in effect size of 0.023 corresponding to an increase in the percentage gain by 0.37%”; a 2024 Bayesian meta-regression by Pelland/Robinson et al. estimated 0.24% per added set at ~12 weekly sets, 95% CrI 0.15–0.33, with diminishing returns), proximity-to-failure matters more for hypertrophy than strength, protein plateaus around 1.62 g/kg, and periodization style barely matters versus simply progressing. Atlas must encode these as *adjustable heuristics with confidence levels*, not laws — MEV/MAV/MRV landmarks, “effective reps,” and readiness scores are all individual estimates, not measured truths.
- **Build in this order (MVP → advanced):** (1) safety red-flags + exercise database + logging; (2) deterministic progression and volume tracking; (3) fatigue/readiness scoring and deload logic; (4) personalization by goal/level; (5) long-term memory, plateau detection, and a confidence system that makes Atlas *ask* rather than assume whenever data is missing or contradictory.

Key Findings

1. **Most “rules” in strength coaching are heuristics with real scientific debate behind them.** Volume landmarks (MEV/MAV/MRV) popularized by Dr. Mike Israetel of Renaissance Periodization are explicitly individual estimates that shift week to week; even RP states “these are directions, not dogmas.” The dose-response between volume

and growth is real but the slope is small and the upper limit contested. Atlas must treat numbers as starting points and personalize from the user's own response data.

2. **The single biggest design decision is the deterministic/LLM split.** Anything that must be consistent, auditable, and safe — load progression, deload triggers, volume sums, fatigue scores, plateau detection, injury red-flags — belongs in code. Anything requiring natural language, empathy, education, or synthesis of messy context belongs to the LLM. The LLM should *call* the rules engine and explain its outputs, never invent its own progression math.
3. **A confidence system is what separates an elite coach from a guessing one.** Elite coaches ask before assuming. Atlas needs an explicit confidence score per decision, driven by data completeness, recency, consistency, and self-report reliability — with thresholds that trigger clarifying questions instead of action.
4. **Readiness/recovery scoring is useful but must be honest about its limits.** HRV and subjective wellness track fatigue imperfectly; the research consensus is that subjective mood/motivation measures are among the *most* reliable early warning signs, while single biomarkers (cortisol, T:C ratio) are unreliable for diagnosing overtraining. Atlas should weight cheap, validated subjective inputs heavily and treat wearable data as supporting, not decisive.
5. **A training coach needs only “minimum viable nutrition”** — protein targets, energy balance, rate-of-change guidance, hydration, and a short evidence-ranked supplement list — plus hard guardrails to refer out for clinical nutrition, eating-disorder signs, or medical conditions.

Details

PART 1 — THE COMPLETE COACHING KNOWLEDGE TAXONOMY (Deliverable 1)

Atlas's knowledge is organized into eleven top-level domains. For each, I give: why it matters, what Atlas must know, what becomes deterministic code, what stays LLM reasoning, required user data, helpful historical data, and what is stored permanently vs inferred at runtime.

DOMAIN A — TRAINING SCIENCE (the principles engine)

Why it matters: This is the “physics” of coaching. Every recommendation traces back to a

handful of principles: specificity, overload, progression, fatigue management, individualization, and the SAID principle (Specific Adaptation to Imposed Demands).

What Atlas should know:

- **Progressive overload** — the gradual increase of stress over time (ACSM's definition: "the gradual increase of stress placed upon the body during exercise training"). Achieved via load, reps, sets, density, ROM, or proximity-to-failure.
- **Hypertrophy** — driven primarily by mechanical tension; total weekly volume (sets per muscle) is the dominant dose variable. Schoenfeld, Ogborn & Krieger (2017, *J Sports Sci* 35:11) found a graded dose-response in which "each additional set was associated with an increase in effect size (ES) of 0.023 corresponding to an increase in the percentage gain by 0.37%"; a 2024 Bayesian meta-regression (Pelland/Robinson et al.) estimated a smaller ~0.24% per added set at ~12 weekly sets (95% CrI 0.15–0.33) with clear diminishing returns.
- **Strength development** — driven by neural adaptation + hypertrophy; favors higher intensity (load), lower reps, longer rest, specificity to the tested lift.
- **Muscular endurance** — higher reps, shorter rest, metabolic conditioning.
- **Power / rate of force development (RFD) / speed work** — high intent/velocity, low-to-moderate load, low fatigue, far from failure; plyometrics and ballistic work. Power exercises (Olympic lifts) are placed first in a session.
- **Athletic performance** — concurrent training, transfer, force-velocity profiling.
- **Periodization** — linear, block, daily undulating (DUP), conjugate. Key honest finding: meta-analyses (Harries et al. 2015; later work) find no consistent superiority of one model over another for strength when volume is equated; the 2009 ACSM stand emphasized periodization more than the contemporary evidence supports. Periodization mostly matters as a *structure for progression and fatigue management*, not a magic variable.
- **Deload / taper / peaking** — planned volume/intensity reductions to dissipate fatigue and realize fitness (the fitness-fatigue model).

Deterministic rules: volume summation per muscle/week; intensity zones (%1RM → rep ranges); progression triggers; periodization scheduling (block/wave templates); deload scheduling; e1RM estimation.

LLM reasoning: explaining *why* a principle applies to this user; resolving trade-offs ("you want size and strength — here's how we balance them"); narrating the plan.

Required user data: goal, training age, current program, available days/equipment.

Historical data that improves decisions: response to past volume/intensity, rate of strength gain, fatigue accumulation patterns. **Permanent vs inferred:** principles are permanent knowledge; the user's *personal dose-response* is inferred and continuously updated.

DOMAIN B — EXERCISE SCIENCE (the movement database) — THE ATTRIBUTE FRAMEWORK

Why it matters: Atlas can't reason about substitutions, fatigue, or technique without a structured model of every movement. Rather than an exhaustive matrix, Atlas needs a **schema** the user populates over time.

The classification backbone (two authoritative systems):

1. **NSCA classification** (from *Essentials of Strength Training and Conditioning*, Haff & Triplett eds.): exercises are **Core vs Assistance**, and core exercises can be **Structural** (loads the spine directly/indirectly) and **Power** (a structural exercise performed very quickly). Core exercises "use one or more large muscle areas, involve two or more joints, and receive priority"; assistance exercises "use smaller muscle areas and one joint."
2. **Movement-pattern framework** (Dan John's five basic human movements — **push, pull, hinge, squat, loaded carry** — popularized in *Intervention*, 2013; commonly extended with **lunge** (Dr. John Rusin's 6 patterns) and **rotation/anti-rotation**). Distinct from this is **Gray Cook & Lee Burton's Functional Movement Screen (FMS, finalized 1997; Cook 2010)** — a *screening* taxonomy of seven patterns for injury-risk assessment, not a programming taxonomy. Atlas should keep these separate: one organizes programming, the other assesses movement quality.

THE EXERCISE ATTRIBUTE SCHEMA (every exercise gets these fields):

Attribute	Type	Example values
name / aliases	text	"Back Squat", "BB Squat"
movement_pattern	enum	squat, hinge, horizontal_push, vertical_push, horizontal_pull, vertical_pull, lunge, carry, rotation, anti_rotation, isolation, locomotion
nsca_class	enum	core_structural_power, core_structural, core, assistance
primary_muscles	list	quadriceps, gluteus maximus

secondary_muscles	list	adductors, erectors
stabilizers	list	trunk, spinal erectors
implement	enum	barbell, dumbbell, machine, cable, kettlebell, bodyweight, band
loading_axis	enum	axial, non-axial
systemic_fatigue	1–5	5 = very high (heavy deadlift)
local_fatigue	1–5	per target muscle
joint_stress	map	knee, low_back, shoulder ratings
technical_complexity	1–5	5 = snatch
stimulus_to_fatigue (SFR)	qualitative	high/medium/low — see caveat
skill_progressions	list	regressions & progressions
common_faults	list	text
substitutions	list	exercise IDs with similarity score
contraindications	list	injury flags
ROM_profile / resistance_profile	text	where it's hardest

A note on stimulus_to_fatigue (SFR): the stimulus-to-fatigue ratio — positive adaptation to a target muscle relative to total fatigue cost — was coined/popularized by Dr. Mike Israetel (Renaissance Periodization, *Scientific Principles of Hypertrophy/Strength Training*). It is a practical heuristic with no objective units; store it as a qualitative tag, not a measured value.

WORKED EXAMPLES:

- **Back Squat** — pattern: squat; NSCA: core/structural (can be power as jump squat); primary: quads, glutes; secondary: adductors, hamstrings (co-contraction); stabilizers: spinal erectors, trunk, upper back; implement: barbell; loading axis: **axial**; systemic fatigue: 4–5; local fatigue: quads 5; joint stress: knee + low back moderate-high; complexity: 4; SFR: high for quads but with substantial axial/systemic cost. Progressions: goblet → front → high-bar → low-bar; regressions: box squat, leg press,

belt squat (removes axial load). Common faults: knee valgus, butt wink, heels rising, excessive forward lean. Substitutions: front squat, hack squat, leg press, belt squat. Injury notes: low-back and knee pathology → shift to belt squat / leg press to keep stimulus, cut axial fatigue.

- **Conventional Deadlift (hinge)** — pattern: hinge; NSCA: core/structural; primary: glutes, hamstrings, erectors; secondary: quads, lats, traps; stabilizers: whole posterior chain, grip; implement: barbell; loading axis: axial; systemic fatigue: **5 (coaching consensus: among the highest)**; local fatigue: posterior chain high; joint stress: low back high; complexity: 4. **Honest caveat:** RP/Israetel rank the deadlift as the most systemically/axially fatiguing lift via the SFR concept, but head-to-head research (Barbalho et al., *JSCR* 2017, squat vs. deadlift) found *no difference in central fatigue* and actually *more peripheral fatigue from the squat*. So Atlas should treat the “deadlift = highest fatigue” rule as a useful default heuristic, flagged as expert opinion, not settled science. Regressions: trap-bar deadlift (less low-back shear), Romanian deadlift, rack pull, hip thrust. Faults: lumbar flexion, hips shooting up, bar drifting forward.
- **Bench Press (horizontal push)** — pattern: horizontal_push; NSCA: core; primary: pec major, anterior delt, triceps; stabilizers: rotator cuff, lats, legs (leg drive); implement: barbell; loading axis: non-axial; systemic fatigue: 3; local fatigue: chest/triceps; joint stress: shoulder (especially with flared elbows / extreme arch); complexity: 3. Regressions: dumbbell press (shoulder-friendly ROM), machine chest press, push-up. Faults: bouncing, uneven lockout, elbow flare, losing upper-back tightness. Substitutions: DB bench, incline press, weighted dip, machine press.
- **Pull-up (vertical pull)** — pattern: vertical_pull; NSCA: core; primary: lats, biceps; secondary: mid-back, rear delt; stabilizers: scapular stabilizers, grip, trunk; implement: bodyweight; systemic fatigue: 2–3; local fatigue: lats/biceps; joint stress: shoulder/elbow moderate; complexity: 3. Regressions: band-assisted, machine assisted, lat pulldown, inverted row. Progressions: weighted pull-up. Faults: partial ROM, excessive kipping when strict is the goal, no scapular engagement.

Deterministic: substitution matching (by pattern + muscles + equipment + fatigue similarity); fatigue summation by muscle and systemically; complexity-gating for beginners. **LLM:** cueing, fault diagnosis from user description, explaining why a substitution preserves the stimulus. **Required user data:** equipment, injury history, experience. **Historical:** which exercises the user tolerates/enjoys/responds to. **Permanent vs inferred:** the exercise database is permanent (a curated asset); the user’s per-exercise tolerance and PRs are stored permanently in their profile; suitability for *today* is inferred.

DOMAIN C — PROGRAMMING (how elite coaches build sessions)

Why it matters: This is where principles become a concrete plan.

What Atlas should know: volume (sets × reps × load), intensity (%1RM / RPE / RIR), frequency, exercise order (power → core → assistance; large → small; multi- → single-joint), warmups (general + specific ramping sets), rest periods (longer for strength/compounds ~2–5 min, shorter for isolation/endurance ~30–90s), and intensity techniques: supersets, giant sets, drop sets, cluster sets, rest-pause, tempo prescriptions, paused reps. Autoregulation tools: **RPE/RIR** (the Zourdos/Helms RIR-based scale: RPE 10 = 0 reps in reserve, RPE 8 = 2 RIR), velocity-based training (VBT) with velocity-loss thresholds, and AMRAP sets for calibration.

Honest uncertainty: The “effective reps”/“stimulating reps” model (Beardsley) — that only the last ~5 reps before failure drive growth — is *contested*. Greg Nuckols/Stronger by Science argue the supporting evidence is weak. Atlas should not hard-code “effective reps” as truth; use proximity-to-failure as the practical lever instead. Robinson, Pelland, Zourdos et al. (2024) found hypertrophy increases as sets approach failure while strength is similar across a wide RIR range — but the authors explicitly caution the analysis is exploratory.

Deterministic: rest-period defaults by goal; set/rep schemes by goal and level; exercise-order sorting; warmup ramp generation; RIR→load conversions; VBT thresholds. **LLM:** explaining technique choices, adjusting language to the user, teaching what RIR feels like. **Required:** goal, days/week, session length, equipment, experience. **Historical:** actual rest taken, RIR accuracy over time, tolerated frequency. **Permanent vs inferred:** templates permanent; the user’s calibrated RIR accuracy stored permanently; today’s session generated at runtime.

DOMAIN D — PROGRESSION INTELLIGENCE (the decision core)

Why it matters: This is the most valuable, most frequent decision Atlas makes and the one that most needs to be deterministic and consistent.

The decision set: increase load / hold load / decrease load / add reps / drop reps / add sets / cut sets / swap exercise / add variation / hold and reassess.

Decision logic (deterministic core, examples):

- **Double progression:** when the user hits the top of the rep range at target RIR for all sets across required sessions, increase load (typically 2.5–5% upper body, 5–10 lb lower body, or smallest available increment). ACSM guidance: apply a 2–10% load increase when the trainee can perform 1–2 reps over the target.
- **Beginner linear progression:** add load every successful session; on a missed session, repeat; on 2–3 consecutive misses, deload ~10% and re-climb.

- **Volume progression within a mesocycle:** start near MEV, add sets toward MAV when recovery is adequate, deload before MRV.
- **One bad session vs. true plateau:** Atlas should **ignore a single underperformance** if isolated and explainable (poor sleep, high stress, missed nutrition). It should flag a **plateau** only when performance fails to progress across a defined window (e.g., ≥ 3 sessions or 2–3 weeks) *despite* adequate recovery markers — and then change a variable (load scheme, volume, exercise, or trigger a deload).

Differential diagnosis of underperformance — Atlas reasons across causes by cross-referencing logged data, not guessing:

Likely cause	Signals Atlas can detect
Poor sleep	self-reported sleep ↓, single session, isolated
Acute fatigue	recent high-volume/intensity, multi-lift dip, recovers in days
Under-recovery (chronic)	multi-week decline, rising session-RPE for same load, low motivation
Injury	localized pain, one movement pattern affected, sharp/asymmetric
Illness	systemic, elevated resting HR, malaise reported
Loss of motivation	adherence ↓, RPE-effort mismatch, skipped sets, mood reports
Bad programming	predictable stall tied to excessive volume/insufficient progression structure

Deterministic: all the above triggers and thresholds. **LLM:** asking the clarifying question that disambiguates cause (“Was this just an off day, or have the last few felt harder than the weight should?”), and delivering the verdict empathetically. **Required:** per-set logs (load, reps, RIR), session RPE, basic wellness check-ins. **Historical:** rolling performance trends, expected-performance baselines. **Permanent vs inferred:** thresholds are configurable constants; the verdict is inferred each time; confirmed plateaus and their resolutions are logged to coach memory permanently.

DOMAIN E — FATIGUE & RECOVERY

Why it matters: Fitness minus fatigue equals performance (the **Banister fitness-fatigue / impulse-response model**, 1975): each training dose creates both a long-lasting fitness gain and a larger, faster-decaying fatigue cost (fatigue typically modeled as decaying $\sim 3\times$

faster than fitness). Coaching is the art of managing the fatigue side.

What Atlas should know: central vs peripheral fatigue; local muscle vs systemic/axial fatigue; muscle-specific recovery (small muscles ~24–48h, large/compound ~48–72h, [Betterlifefitness](#) very heavy eccentric or to-failure work up to 72h+ — all approximate and individual); connective-tissue lags behind muscle; and modulators: sleep, nutrition/energy availability, psychological stress, illness, travel, heat, age, training age, and concurrent (endurance) training.

Overtraining continuum: functional overreaching (planned, 1–2 wk, supercompensates) → non-functional overreaching (3–4 wk decline, no supercompensation) → overtraining syndrome (weeks-to-months). The ECSS/ACSM consensus (Meeusen et al. 2013) emphasizes that training is not the only stressor and that subjective markers (mood, motivation, perceived exertion mismatched to load) are among the most reliable early signs, while no single biomarker reliably diagnoses OTS.

Readiness — and its honest limits: Atlas can compute a composite readiness score, but the literature is clear that **subjective wellness measures (mood, motivation, perceived fatigue, soreness) are among the most reliable early indicators**, [Getfitcraft](#) while HRV is noisy day-to-day ("high isn't always good, low isn't always bad") [SimpliFaster](#) and single biomarkers are unreliable. So Atlas should weight cheap subjective check-ins heavily and treat wearables as supporting evidence with wide error bars.

Deterministic: per-muscle recovery clocks; rolling acute:chronic load ratio; readiness composite (weighted: sleep, soreness, stress, motivation, recent load, optional HRV); deload triggers (planned every ~4–6 weeks OR auto-triggered by sustained readiness decline / performance drop / elevated session-RPE). **LLM:** interpreting the readiness picture in plain language; deciding when a low score warrants a conversation vs. an automatic adjustment. **Required:** sleep, soreness, stress, motivation check-ins; session load.

Historical: individual recovery rates, deload responses, baselines. **Permanent vs inferred:** baselines and recovery-rate estimates stored permanently; today's readiness inferred.

DOMAIN F — PERSONALIZATION

Why it matters: The same principle is applied differently to a 19-year-old powerlifter and a 55-year-old busy parent.

Dimensions Atlas must adapt to:

- **Training age:** beginners progress every session (linear), need lower complexity and volume, faster gains; intermediates need more variation and structured progression; advanced need higher volume, periodization, specialization, and recover more slowly

relative to stimulus. (ACSM frequency guidance: novice 2–3 d/wk, intermediate 3–4, advanced 4–5+.) [Medscape](#)

- **Goal:** powerlifting (specificity, intensity, peaking), bodybuilding (volume, muscle balance, proximity-to-failure), general fitness (minimum effective dose, sustainability), weight loss (preserve muscle, manage fatigue in a deficit), athletic performance (power, concurrent training, transfer).
- **Populations:** older adults (more warmup, joint-friendly choices, protein emphasis, longer recovery, but still progressive and intense enough to matter); youth (technique, supervision, conservative loading); busy parents (time-efficient, autoregulated, low-guilt missed-session handling); limited equipment / home vs commercial gym (substitution-driven programming).

Deterministic: template selection by level/goal; volume/frequency/complexity caps by population; equipment-constrained exercise filtering. **LLM:** tone, framing, and lifestyle-sensitive coaching. **Required:** age, training age, goal, equipment, schedule, constraints. **Historical:** adherence patterns, life-context changes. **Permanent vs inferred:** profile attributes permanent; “current life season” (e.g., new baby, injury rehab) is a permanent-but-updatable state.

DOMAIN G — BEHAVIOR CHANGE & COACHING PSYCHOLOGY

Why it matters: The best program is the one the user actually does. Adherence is the master variable.

What Atlas should know:

- **Habit formation:** Lally, van Jaarsveld, Potts & Wardle (2010, *Eur J Soc Psychol* 40:998–1009) found “the time it took participants to reach 95% of their asymptote of automaticity ranged from 18 to 254 days” (median 66 days), and crucially that “missing one opportunity to perform the behaviour did not materially affect the habit formation process” — an anti-guilt message Atlas should encode directly. Context-cue consistency matters more than streak perfection.
- **Self-Determination Theory** (Deci & Ryan): support **autonomy** (offer choices, give rationale), **competence** (achievable goals, specific feedback, mastery), and **relatedness** (feeling seen by the coach). [University of Rochester Medical Center](#) Autonomous motivation predicts long-term adherence far better than external pressure. [University of Rochester Medical Center](#)
- Identity-based change (“you’re becoming a person who trains”), self-efficacy, goal-

setting (process > outcome goals), and accountability without nagging.

Avoiding repetitiveness: Atlas should rotate encouragement styles, reference *specific* user history ("your squat's up 15 kg since March"), vary cadence, and *earn* praise with specifics rather than generic "great job!"

Deterministic: adherence tracking; streak/consistency metrics; nudge timing rules; detection of declining engagement. **LLM:** all motivational language, reframing setbacks, identity reinforcement, choosing what to say and when to stay quiet. **Required:** adherence logs, stated goals, occasional mood check-ins. **Historical:** what messaging worked, drop-off patterns, preferences. **Permanent vs inferred:** motivational profile and preferences permanent; today's emotional state inferred.

DOMAIN H — COMMUNICATION

Why it matters: Communication quality *is* the product for an AI coach.

Principles Atlas should embody: ask intelligent questions before assuming; admit uncertainty honestly ("the evidence here is mixed"); always explain *why*; reference user history; avoid generic praise and motivational clichés; recognize long-term patterns; be encouraging without exaggeration; calibrate verbosity to the user.

Deterministic: none of the language itself; but deterministic *triggers* decide when to surface a question (driven by the confidence system) or a milestone. **LLM:** essentially all of it. **Permanent vs inferred:** communication preferences (length, tone, technical depth) stored permanently.

DOMAIN I — NUTRITION (minimum viable for a training coach)

Why it matters: Training adaptation is gated by protein and energy. A training coach must know enough to support the program and refer out for the rest.

The minimum knowledge set:

- **Protein:** ISSN Position Stand (Jäger et al. 2017, *J Int Soc Sports Nutr* 14:20): "an overall daily protein intake in the range of 1.4–2.0 g protein/kg body weight/day (g/kg/d) is sufficient for most exercising individuals," rising to 2.3–3.1 g/kg/d to retain lean mass during hypocaloric periods. The practical optimum comes from Morton et al. 2018 (*Br J Sports Med* 52:376–384), whose biphasic regression found "a plateau in RET-induced gains in FFM at 1.62 g protein/kg/day (95% CI: 1.03 to 2.20)," with the authors suggesting up to ~2.2 g/kg to maximize gains. Aim ~0.4 g/kg per meal across 3–4 meals;

total daily intake matters most.

- **Energy balance:** surplus for maximal muscle gain, deficit for fat loss, maintenance for recomposition/performance.
- **Rate of change:** lose ~**0.5–1% of bodyweight/week** to preserve muscle (Murphy & Koehler 2021 found lean-mass loss rises as deficits exceed ~500 kcal/day); lean bulk slowly.
- **Bulking/cutting/maintenance** logic; **meal timing** (minor vs total intake — protein around training is helpful but secondary); **hydration**; **recovery nutrition** (protein + carbs).
- **Supplements (evidence-ranked):** creatine monohydrate — per ISSN (Kreider et al. 2017, *J Int Soc Sports Nutr* 14:18), "the most effective ergogenic nutritional supplement currently available to athletes," at "3–5 g/day or 0.1 g/kg of body mass/day"; caffeine (3–6 mg/kg pre-training); beta-alanine (4–6 g/day for buffering, per ISSN). Most others weak/unnecessary. Protein powder = convenience, not magic.

Deterministic: protein target calculator (g/kg); calorie estimate from bodyweight/goal; rate-of-change guardrails; supplement dosing reference. **LLM:** explaining, answering food questions, adapting to preferences/diets. **HARD GUARDRAIL:** Atlas does *not* prescribe clinical diets, treat medical conditions, or counsel on disordered eating — it refers to a registered dietitian/physician and flags eating-disorder warning signs. **Required:** bodyweight, goal; optional dietary preferences. **Permanent vs inferred:** targets stored and recomputed as bodyweight changes.

DOMAIN J — INJURY & SAFETY (the guardian layer)

Why it matters: This is the domain where errors are catastrophic, so it is *maximally deterministic and conservative*.

The escalation ladder:

1. **Coach normally** — no pain, normal soreness (DOMS), good readiness.
2. **Modify / reduce intensity** — minor joint niggle, low readiness, mild localized discomfort that doesn't worsen → reduce load, swap to pain-free variation, cut volume.
3. **Stop the session** — pain that is sharp, worsening, or alters movement; dizziness; unusual breathlessness.
4. **Refer to medical evaluation, do not coach through** — RED FLAGS: chest pain/pressure/tightness (especially with radiation to arm/jaw, shortness of breath,

nausea, cold sweats); [SiPhox Health](#) fainting or near-fainting; [U.S. News & World Report](#) sudden severe headache; numbness/weakness/tingling; sharp or “tearing” pain; [Liv Hospital](#) joint instability or suspected acute injury; symptoms that persist after stopping. These map to clinical “absolute indications to stop exercise” (e.g., suspected cardiac event, moderate-to-severe angina, signs of poor perfusion, CNS symptoms like ataxia/vertigo/confusion).

Safe defaults: when uncertain, Atlas always errs toward caution; it never diagnoses; it documents and recommends professional evaluation. It uses a PAR-Q-style screen at onboarding.

Deterministic: red-flag symptom classifier (keyword/structured-input triggers → immediate stop/refer message); contraindication filtering against injury profile; load-reduction rules.

LLM: empathetic delivery of the stop/refer message; asking about pain quality to classify it.

Required: injury history, health screen, in-session pain/symptom reports. **Permanent vs inferred:** injuries and contraindications stored permanently; today’s symptom interpretation inferred but always conservative.

DOMAIN K — KNOWLEDGE SOURCES (ranked by trustworthiness) (part of Deliverable 1)

Atlas should weight its knowledge base by evidence quality. Ranked tiers:

Tier 1 — Consensus statements & position stands (highest trust for practice guidance): ACSM (progression models; resistance training prescription), NSCA, ISSN (protein, creatine, beta-alanine, diets), IUSCA. *Best for:* defensible default recommendations. *Caution:* can lag the newest research; the 2009 ACSM periodization emphasis is an example of guidance the literature later softened.

Tier 2 — Systematic reviews & meta-analyses: Schoenfeld/Krieger (volume, frequency), Morton (protein), Refalo/Robinson/Zourdos (proximity-to-failure), Harries (periodization). *Best for:* dose-response and effect sizes. *Caution:* heterogeneity, mostly young trained males, publication bias, exploratory meta-regressions (Robinson 2024 explicitly cautions on interpretation). [Sportrxiv](#)

Tier 3 — Major textbooks: NSCA *Essentials of Strength Training and Conditioning* (Haff & Triplett); Zatsiorsky & Kraemer *Science and Practice of Strength Training*; Siff/Verkhoshansky *Supertraining*. *Best for:* foundational frameworks, classification, periodization theory. *Caution:* *Supertraining* in particular contains theory ahead of (or without) strong evidence.

Tier 4 — Leading researchers (primary literature & evidence reviews): Brad Schoenfeld

(hypertrophy), Stuart Phillips (protein), Eric Helms (RPE/autoregulation, natural bodybuilding), Greg Nuckols (Stronger by Science, MASS), Mike Zourdos (RIR), James Krieger. *Best for:* nuance and current debates. *Caution:* individual studies aren't consensus.

Tier 5 — Evidence-based practitioners / orgs: Renaissance Periodization (Israetel — volume landmarks, SFR), Stronger by Science, 3DMJ, MASS Research Review, reputable university S&C programs. *Best for:* practical frameworks and application. *Caution:* RP's MEV/MAV/MRV and SFR are *heuristics/models*, not validated measurements — flag them as such.

Lowest trust: anecdote, influencer marketing, single-study sensationalism, supplement-company claims. Atlas should down-weight or exclude these.

PART 2 — PRIORITIZED IMPLEMENTATION ROADMAP (Deliverable 2)

Phase 0 — Foundations (MVP-critical):

- Safety/red-flag classifier + onboarding health screen (build first; it's a guardrail).
- Exercise database with the attribute schema (seed ~50–100 common exercises).
- Workout logging (load, reps, RIR, session RPE).
- Permanent user profile (goal, training age, equipment, injuries).

Phase 1 — Core coaching brain (MVP):

- Deterministic progression engine (double progression + beginner LP).
- Volume tracking per muscle/week against MEV/MAV/MRV defaults.
- e1RM estimation and basic trend lines.
- LLM layer for explanation + encouragement (constrained to call the engine).

Phase 2 — Recovery & autoregulation:

- Readiness scoring (subjective-weighted) and per-muscle recovery clocks.
- Deload logic (planned + auto-triggered).
- Plateau detection and the underperformance differential.

Phase 3 — Personalization depth:

- Goal/population-specific templates and caps.
- Nutrition module (protein/calorie/rate guardrails + supplement reference).

- Adherence/behavior-change nudging.

Phase 4 — Advanced intelligence:

- Long-term trend memory and individualized dose-response learning.
 - Confidence system maturity (asks vs. acts).
 - Periodization scheduling, specialization blocks, VBT support.
 - Coach memory that references long-term patterns conversationally.
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PART 3 — COACHING KNOWLEDGE BASE STRUCTURE (Deliverable 3)

Organize stored knowledge into **three forms**, by use:

1. **Structured rules & tables (the source of truth for decisions):** exercise database; volume-landmark tables by muscle/level; intensity→rep-range maps; RIR→%1RM tables; progression thresholds; recovery-rate constants; red-flag symptom table; supplement dosing table; nutrition target formulas. These are versioned, human-readable config (not buried in code) so the non-coder owner can tune them.
2. **Prose knowledge for retrieval (RAG):** explanations of principles, the “why” behind rules, technique cues, common-fault libraries, FAQ-style education, and honest write-ups of scientific debates (effective reps, volume ceilings, readiness limits). Chunked, tagged by domain, with source + evidence-tier metadata so the LLM can cite quality.
3. **Decision provenance / rationale store:** for each rule, a short note on *why it exists, its source, its evidence tier, and its confidence*. This lets the LLM explain decisions accurately and lets the owner audit and update.

Principle: numbers and decisions live in structured form; understanding and language live in prose. Never let the LLM regenerate a number that a table should own.

PART 4 — DETERMINISTIC RULES ENGINE STRUCTURE (Deliverable 4)

Modular engine; each module owns one class of decision and is independently testable:

- **SafetyModule** — red-flag classification, contraindication filtering, stop/refer triggers. *Highest priority; can veto any other module.*
- **VolumeModule** — sums sets per muscle/week; compares to MEV/MAV/MRV; flags under/over-reaching.

- **ProgressionModule** — load/rep/set decisions (double progression, LP, autoregulated).
- **IntensityModule** — RIR/RPE↔load, %1RM zones, e1RM.
- **FatigueModule** — per-muscle recovery clocks, acute:chronic load, systemic-fatigue accumulation.
- **ReadinessModule** — composite readiness score from weighted inputs.
- **DeloadModule** — planned + auto-triggered deload scheduling.
- **PlateauModule** — trend detection, plateau vs. noise, underperformance differential.
- **SubstitutionModule** — exercise swaps by pattern/muscle/equipment/fatigue similarity.
- **ExpectedPerformanceModule** — predicts today's likely loads/ reps as a baseline to compare against.
- **NutritionModule** — protein/calorie targets, rate-of-change guardrails.
- **ConfidenceModule** — scores confidence and decides ask-vs-act (cross-cuts all others).

Modules output structured decisions + a rationale + a confidence value; the LLM consumes these.

PART 5 — LLM PROMPT ARCHITECTURE (Deliverable 5)

Layered system prompt (assembled per turn):

1. **Identity & guardrails layer (static):** Atlas's role, coaching philosophy, hard rules ("never override the rules engine; never diagnose; refer on red flags; admit uncertainty; explain why; don't fabricate numbers").
2. **Knowledge layer (semi-static):** distilled coaching principles + evidence tiers; instructions to cite quality and flag debates.
3. **User-context layer (dynamic, injected per turn):** permanent profile, current goal, injuries, preferences, recent trends, last sessions — pulled from memory.
4. **Engine-output layer (dynamic):** the deterministic decisions + rationale + confidence for *this* interaction.
5. **Conversation layer:** recent dialogue + the user's latest message.

Tool / function-calling interface: the LLM does not compute coaching math; it *calls functions* — `get_progression_decision()` , `get_readiness()` , `get_volume_status()` ,

`find_substitution()` , `log_workout()` , `check_safety()` , `get_nutrition_targets()` . It receives structured results and translates them into coaching language.

How knowledge is injected: structured decisions are passed directly as data; prose education is retrieved via RAG and grounded; the LLM is instructed to prefer engine outputs and retrieved sources over its own parametric memory for any factual/numeric claim.

How the LLM is constrained: explicit prohibition on inventing loads/sets/percentages; required to defer to `SafetyModule` ; required to express uncertainty when the engine's confidence is low; required to ask a clarifying question when the `ConfidenceModule` says so.

PART 6 — MEMORY ARCHITECTURE (Deliverable 6)

Four memory stores, separated by lifespan and stability:

1. **Permanent Profile (slow-changing, authoritative):** identity, age, training age, goals, equipment, injuries/contraindications, communication & motivational preferences, dietary constraints. Stored permanently; updated deliberately.
2. **Long-Term Trends (derived, periodically recomputed):** strength trajectory per lift, e1RM history, individual dose-response (what volume/intensity the user responds to), recovery-rate estimates, adherence patterns, plateau/deload history. *Stored* as rolling summaries; the underlying detail is in history; trends are recomputed, not free-form remembered.
3. **Session Memory (current workout):** today's planned vs. actual sets, in-session RIR/RPE, symptoms, readiness. Promoted into history at session end.
4. **Conversation Memory (dialogue):** recent turns verbatim (short window) + a running summary of older conversation (preferences expressed, questions asked, emotional context). Key facts surfaced in conversation get *promoted* to the permanent profile.

Stored vs inferred: Store raw logs, profile, and confirmed events permanently (cheap, auditable, ground truth). *Infer* at runtime: readiness, today's suitability, expected performance, current emotional state, plateau status. Rule of thumb — **store facts and decisions; infer judgments.**

PART 7 — CONFIDENCE SYSTEM (Deliverable 7)

Why: An elite coach asks rather than assumes. Atlas needs to know what it doesn't know.

Confidence score (0–1) per decision, from four inputs:

- **Data completeness** — are required inputs present? (missing sleep, RIR, or recent logs lowers it).
- **Data recency** — how stale is the data? (a 3-week gap lowers confidence in baselines).
- **Data consistency** — do signals agree? (good logs but contradictory wellness reports lower it).
- **Self-report reliability** — calibrated RIR accuracy, history of over/under-reporting.

Thresholds (tunable):

- **High (≥ 0.8): act.** Make the recommendation, explain briefly.
- **Medium (0.5–0.8): act but verify.** Recommend, but state the assumption and invite correction ("Assuming last week's weights still feel right — adjust me if not").
- **Low (< 0.5): ask before acting.** Trigger a clarifying question rather than guessing.

How missing/low-quality data lowers confidence: each module reports both a decision and a confidence; the `ConfidenceModule` aggregates and gates the LLM. Critically, **the SafetyModule inverts this logic** — low confidence about a possible red flag triggers *more* caution and referral, never less.

This is also the antidote to generic coaching: because Atlas only asks when it genuinely lacks information (not reflexively), its questions feel intelligent and earned — exactly how an elite human coach operates.

Recommendations

Start here (next 90 days):

1. Build the `SafetyModule` and onboarding health screen *first*. It's the cheapest insurance and the one place errors are unacceptable.
2. Build the exercise database with the full attribute schema; seed 50–100 movements with the four worked examples as templates.
3. Implement deterministic logging + double-progression + per-muscle weekly volume tracking. This alone is a competent coach.
4. Add a thin, heavily-constrained LLM layer that *only* explains engine outputs and encourages — prove it never invents numbers.

Then (months 3–9): readiness scoring (subjective-weighted), deload logic, plateau detection + underperformance differential, and the confidence system. These turn a tracker into a coach.

Then (months 9+): personalization templates, nutrition module, behavior-change nudging, long-term trend learning, and conversational coach memory.

Benchmarks that should change your plan:

- If users frequently report the AI “feels generic,” prioritize the confidence system and coach memory (referencing specific history) over new training features.
- If progression recommendations feel wrong, your RIR data is probably unreliable — invest in RIR calibration and weight objective load/rep data higher.
- If adherence is the problem (not programming), shift resources from training-science depth to behavior-change and communication.
- If you ever consider letting the LLM make progression or safety decisions directly to save effort — don’t. That’s the one architectural line that should never move.

Tuning philosophy: every number (volume landmarks, recovery clocks, progression increments, confidence thresholds) should be an editable config value, not hard-coded — because they’re heuristics, and the user’s own data will eventually beat the population defaults.

Caveats

- **The science is principled but imprecise.** Volume landmarks (MEV/MAV/MRV), stimulus-to-fatigue ratios, and “effective reps” are *models and heuristics*, not measured quantities — there are no validated units of “fatigue” or “stimulus.” Atlas should present them as useful defaults, personalize aggressively from individual response, and never imply false precision.
- **Readiness scoring has real limits.** Subjective measures outperform single biomarkers, HRV is noisy, and no algorithm reliably distinguishes functional overreaching from early overtraining in real time. Treat readiness as a soft signal, not a verdict.
- **Most research skews to young, trained men.** Recommendations for older adults, women, and beginners carry more uncertainty; Atlas should widen its confidence bands and personalize faster for these groups.
- **Conflicting evidence is the norm, not the exception** (e.g., deadlift fatigue ranking, optimal volume ceilings, periodization superiority). Atlas’s credibility depends on *naming*

these debates rather than papering over them.

- **Atlas is a coach, not a clinician.** It must never diagnose injuries or medical conditions, prescribe clinical nutrition, or coach through red-flag symptoms. The referral guardrail is non-negotiable.
- This document is the coaching *intelligence* architecture; it deliberately excludes the token/build math covered separately, though the two must integrate.